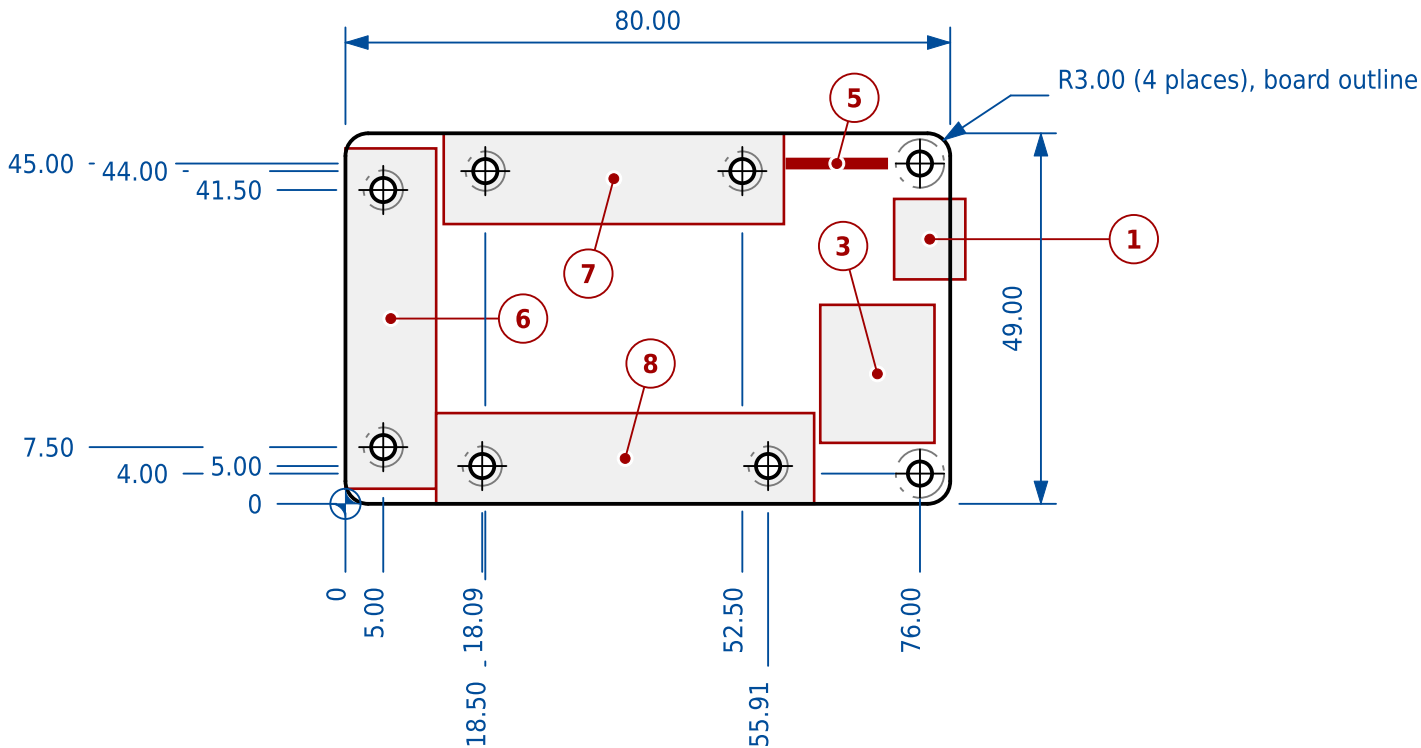




## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| H1  | 76.00 | 45.00 | 3.20   | 6.40       |
| H2  | 76.00 | 4.00  | 3.20   | 6.40       |
| SA1 | 5.00  | 7.50  | 3.20   | 5.20       |
| SA2 | 5.00  | 41.50 | 3.20   | 5.20       |
| SB1 | 18.50 | 44.00 | 3.20   | 5.20       |
| SB2 | 52.50 | 44.00 | 3.20   | 5.20       |
| SC1 | 18.09 | 5.00  | 3.20   | 5.20       |
| SC2 | 55.91 | 5.00  | 3.20   | 5.20       |

## FEATURE SCHEDULE

| # | FEATURE                                      | X EXTENT mm    | Y EXTENT mm    |
|---|----------------------------------------------|----------------|----------------|
| 1 | USB-C J1, programming and console            | 72.58 to 82.00 | 29.68 to 40.32 |
| 2 | Second USB port - not on this board          | -              | -              |
| 3 | Ethernet RJ45 CON1                           | 62.81 to 77.92 | 8.06 to 26.32  |
| 4 | User LEDs, single colour - not on this board | -              | -              |
| 5 | RGB LEDs D1-D7, 7 on 2.00 mm pitch           | 58.40 to 71.60 | 44.40 to 45.60 |
| 6 | SYZGY port A, J3                             | 0.00 to 12.00  | 2.00 to 47.00  |
| 7 | SYZGY port B, J4                             | 13.00 to 58.00 | 37.00 to 49.00 |
| 8 | SYZGY port C, J5                             | 12.00 to 62.00 | 0.00 to 12.00  |

## LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

## NOTES

- Viewed from the component side.
- Hole IDs H1 and H2 are the KiCad reference designators; SA1 to SC2 are the SYZGY standoff holes of ports A, B and C, plated, and the maker's acrylic plate bolts through all eight.
- No Pmod host: expansion is three SYZGY ports.
- Assembled envelope, connector overhang included: 82.00 x 49.00 mm.

## SOURCES

- KiCad board file:  
[https://github.com/butterstick-fpga/butterstick-hardware/blob/main/ButterStick\\_r1.0/ButterStick.kicad\\_pcb](https://github.com/butterstick-fpga/butterstick-hardware/blob/main/ButterStick_r1.0/ButterStick.kicad_pcb) @ fac6a834 - title block: ButterStick rev r1.0a, dated 2021-01-10; Greg Davill / GsD
- Board size:  
<https://github.com/butterstick-fpga/butterstick-hardware/blob/main/README.md> - quoted: "Board dimensions: 80mm x 49mm"
- Thickness:  
[https://github.com/butterstick-fpga/butterstick-hardware/blob/main/hardware/ButterStick\\_r1.0/Production/ButterStick-r1.0a-fab-notes.txt](https://github.com/butterstick-fpga/butterstick-hardware/blob/main/hardware/ButterStick_r1.0/Production/ButterStick-r1.0a-fab-notes.txt) - quoted: "Finished board thickness is 1.6 mm."

## BUTTERSTICK

|                  |        |                                                |       |
|------------------|--------|------------------------------------------------|-------|
| r1.0             |        |                                                |       |
| DRAWN            |        | VERSION                                        | SCALE |
| generated        |        | v0.0-160-g6011fbd                              | 1:1   |
| SIZE             | SHEET  | DRAWING NO                                     | REV   |
| A3               | 1 OF 1 | FPGA-BUTTERSTICK                               | A     |
| MATERIAL         |        | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |        | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |